

A Design Thinking Approach to Construct a Multi-learner VR Lab Monitoring and Assessment Tool Deployed in an XR Environment

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Figure 1: (a) The XR classroom with a large panoramic silver screen and a passive stereoscopic projection system. (b) The control interface of the smartphone-based pointing remote.

ABSTRACT

A Virtual Reality Laboratory (VR Lab) refers to a virtual experiment session that aims to deliver procedural knowledge to students like that in physical laboratories using Virtual Reality (VR) technology. While existing VR Lab designs provide a rich learning experience to students, instructors receive limited information on the student's performance and could not provide feedback and assess their performance. This motivated us to create an XR-based multi-learner monitoring and assessment tool for VR Labs. We evaluated the tool with domain experts and report recommendations for developing monitoring and assessing tools for VR Labs sessions.

Index Terms: Applied computing—Education—Computer-assisted instruction; Applied computing—Education—Interactive learning environments;

1 INTRODUCTION

A virtual reality laboratory (VR Lab) refers to a simulated experiment session being conducted in the virtual environment using virtual reality (VR) technology. The usage of VR labs intends to deliver procedural knowledge like that in real-world laboratories or industrial environments for delivering complementary training materials in addition to those available in the physical world.

There are different research works on VR education. One example is Gao et al. [1], which studied how avatar positioning and visualization affect learning. However, these studies focus on a traditional classroom lecture instead of an interactive laboratory session. XR-LIVE by Thanyadit et al. [4] proposed a system for effective learning of procedural knowledge in a laboratory setting but the design focus on a learner's perspective. VR-MOOCs by Kim et al. [2] proposed a system for monitoring students performing a VR Lab but provides limited progress-tracking and assessment features. This motivated us to create a multi-learner VR Lab monitoring and assessment tool and deployed it in an XR environment.

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2 METHODOLOGY

2.1 A design thinking approach

We adopted the Hasso Plattner Institute Design Thinking approach [3] to create the tool, which includes the stages of empathizing with 10 instructors, defining the design requirements, ideating ideas for meeting the design requirements, prototyping the tool, and testing the tool with experts. A total of 6 requirements are defined after empathizing with the instructors. 4 of them are monitoring requirements which include (R1) - In-depth observation with overview monitoring, (R2) - Showing supplementary information, (R3) - Enable interaction between instructor and students, and (R4) - Adjusting the monitoring scope. The remaining 2 requirements are grading requirements which include (R5) - Selective grading of the tasks and (R6) - Allowing different grading methods. The resulting prototype of the tool is described in Section 3.

2.2 The Unconfined Compression Test

The Unconfined Compression Test (UCT) is a laboratory experiment in civil engineering that aims to test the maximum compressive stress of a specimen. While there are other choices of experiments, we choose the UCT with 9 tasks based on the following criteria:

1. The lab experiment can be divided into multiple tasks and form a learning sequence for learners to obtain procedural knowledge, as well as for the instructor to monitor and assess in stages.
2. The divided tasks are of different operation types that are carried out in a general lab experiment, including equipment placement tasks, measurement tasks, substance mixing tasks, and equipment operation tasks (Figure 2 (a) - (d)).
3. The lab experiment contains the usage of equipment that are usually not available to all the learners at the same time in a physical laboratory, which can demonstrate one of the benefits of conducting the experiment in the virtual world.

In this work, we used VR recordings of 20 students on the UCT for quick prototyping of the tool. A VR recording is created by logging the 3D transformations of the learner's avatar as well as the virtual objects. All 20 recordings originated from [4].

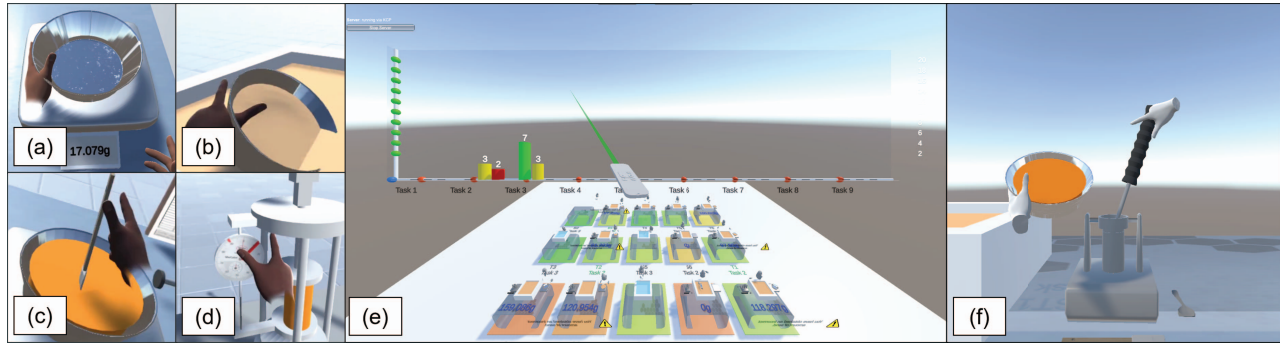


Figure 2: (a) - (d) shows in order an example of the equipment placement task, measurement task, substance mixing tasks and equipment operation tasks in the UCT. (e) shows the condensed working stations of the students, the progress bar chart, as well as the virtual pointing ray. (f) shows an example of the first-person view for manual assessment of a complex task involving hand motions in the UCT.

3 THE PROTOTYPED VR LAB MONITORING AND ASSESSMENT TOOL

3.1 Deployment of the tool in an XR classroom

The prototyped tool is deployed in an XR classroom with a large panoramic silver screen and a passive stereoscopic projection system (**Figure 1(a)**). The projection system is powered by a PC with the configuration (CPU: i7-9700K, RAM: 16GB, Graphics card: NVIDIA Quadro RTX5000) and the rendered stereoscopic graphics can be viewed through polarized glasses. Our tool is written by the Unity game engine and is run on the PC with a stereoscopic interface.

3.2 Interface and interaction model of the tool

The main interface of the tool shows 20 learners participating in a UCT VR Lab. Each student is arranged together with their lab working stations in a 4 x 5 grid layout. Instead of the full working station, a condensed working station, which only consists of the student's avatar and the equipment of a task, is shown (**Figure 2(e)**). A smartphone-based remote pointing device is used to interact remotely with the stereoscopic display. The remote is implemented as an Android application written by the Unity game engine which utilizes the gyroscope data of the Android phone to control a pointing ray for point-and-click interactions (**Figure 2(e)**). The remote also has a control interface (**Figure 1(b)**) with buttons for navigation actions, monitoring actions, and grading actions as well as UI components for text-based interactions. A set of playback control buttons are also provided to control the student recordings for evaluation of the tool.

3.3 Operation of the tool

The learner's progress for a task is color-coded by a tile beneath their workstation. A progress bar chart (**Figure 2(e)**) is used to show an overview of the class's progress and the instructor can filter a single task for monitoring. For detailed monitoring, the instructor can use the remote to point and click on a particular student's condensed lab workstation and immerse into the student's full lab working station. The instructor can freely navigate in the immersed view by moving forward and backward and turning to the remote's pointing direction. Text-based instructions can be attached to the pointed virtual objects for giving hints to the learners. Also, when in an immersed view, a minified main view is shown on the top right-hand corner for simultaneous monitoring of the class. As for grading, a grading policy file is used to specify whether a specific task is non-graded, auto-graded, or manually graded. Auto grading is implemented on object placement tasks and measurement tasks. A prompt message will show up in the student's virtual

environment when the auto-grader grades the answer as incorrect. For manual grading, a student who wishes to demonstrate their task can indicate to the instructor by triggering a blinking animation of their workstation tile. The instructor can then immerse into the student's environment to observe the demo by navigating to a preferred position or switching to a first-person view using the remote (**Figure 2(f)**).

4 EVALUATION

We evaluated the tool by conducting expert reviews with two experts in the field of Civil Engineering. Both experts think that features that satisfy requirements (**R1 – R6**) are useful for monitoring and assessing a live VR Lab session with multiple learners. As recommendations, they suggest including multiple facilitators with different roles for handling larger class sizes and sending notifications to students before immersing into their environment to increase the transparency of the tool.

5 CONCLUSION

In summary, we created an XR-based multi-learner monitoring and assessment tool for VR Labs, using a design thinking approach. The tool is deployed in an XR classroom and controlled by a smartphone-based pointer remote. We evaluated the tool with domain experts and reported recommendations for future developments.

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